TPO 44

[Reading]

1. From Fish to Terrestrial Vertebrates

One of the most significant evolutionary events that occurred on Earth was the transition of water-dwelling fish to terrestrial tetrapods (four-limbed organisms with backbones). Fish probably originated in the oceans, and our first records of them are in marine rocks. However, by the Devonian Period (408 million to 362 million years ago), they had radiated into almost all available aquatic habitats, including freshwater settings. One of the groups whose fossils are especially common in rocks deposited in fresh water is the lobe-finned fish.

The freshwater Devonian lobe-finned fish rhipidistian crossopterygian is of particular interest to biologists studying tetrapod evolution. These fish lived in river channels and lakes on large deltas. The delta rocks in which these fossils are found are commonly red due to oxidized iron minerals, indicating that the deltas formed in a climate that had alternate wet and dry periods. If there were periods of drought, any adaptations allowing the fish to survive the dry conditions would have been advantageous. In these rhipidistians， several such adaptations existed. It is known that they had lungs as well as gills for breathing. Cross sections cut through some of the fossils reveal that the mud filling the interior of the carcass differed in consistency and texture depending on its location inside the fish. These differences suggest a sadlike cavity below the front end of the gut that can only be interpreted as a lung. Gills were undoubtedly the main source of oxygen for these fish, but the lungs served as an auxiliary breathing device for gulping air when the water became oxygen depleted, such as during extended periods of drought. So, these fish had already evolved one of the prime requisites for living on land: the ability to use air as a source of oxygen.

A second adaptation of these fish was in the structure of the lobe fins. The fins were thick, fleshy, and quite sturdy, with a median axis of bone down the center. They could have been used as feeble locomotor devices on land, perhaps good enough to allow a fish to flop its way from one pool of water that was almost dry to an adjacent pond that had enough water and oxygen for survival. These fins eventually changed into short, stubby legs. The bones of the fins of a Devonian rhipidistian exactly match in number and position the limb bones of the earliest known tetrapods, the amphibians. It should be emphasized that the evolution of lungs and limbs was in no sense an anticipation of future life on land. These adaptations developed because they helped fish to survive in their existing aquatic environment.

What ecological pressures might have caused fishes to gradually abandon their watery habitat and become increasingly land-dwelling creatures? Changes in climate during the Devonian may have had something to do with this if freshwater areas became progressively more restricted. Another impetus may have been new sources of food. The edges of ponds and streams surely had scattered dead fish and other water-dwelling creatures. In addition, plants had emerged into terrestrial habitats in areas near streams and ponds, and crabs and other arthropods were also members of this earliest terrestrial community. Thus, by the Devonian the land habitat marginal to freshwater was probably a rich source of protein that could be exploited by an animal that could easily climb out of water. Evidence from teeth suggests that these earliest tetrapods did not utilize land plants as food; they were presumably carnivorous and had not developed the ability to feed on plants.

How did the first tetrapods make the transition to a terrestrial habitat? Like early land plants such as rhyniophytes, they made only a partial transition; they were still quite tied to water. However, many problems that faced early land plants were not applicable to the first tetrapods The ancestors of these animals already had a circulation system, and they were mobile, so that they could move to water to drink. Furthermore, they already had lungs, which rhipidistians presumably used for auxiliary breathing. The principal changes for the earliest tetrapods were in the skeletal system—changes in the bones of the fins, the vertebral column, pelvic girdle, and pectoral girdle.

1. Paragraph 1 supports which of the following statements about fish evolution?
A. Lobe-finned fish were among the earliest types of fish to appear.
B. Fish began living in freshwater habitats only after originating elsewhere.
C. Lobe-finned fish radiated into almost all available aquatic habitats.
D. During the Devonian, lobe-finned fish were more common in marine than in freshwater habitats.

2. According to paragraph 2, what do the minerals in the delta rocks containing rhipidistian crossopterygian fossils reveal?
A. These deltas formed in dry periods but gradually became wetter.
B.These deltas contain different types of iron minerals than do the surrounding areas.
C.Most rhipidistian crossopterygian fish died when the climate became dry.
D.Rhipidistian crossopterygian fish lived in areas that experienced alternate dry and wet periods.

3. The word "advantageous" in the passage is closest in meaning to
A. beneficial
B. necessary
C. remarkable
D. common

4. In paragraph 2, why does the author include the information that mud inside rhipidistian crossopterygian fossils differed in consistency and texture depending on where the mud was located?
A. To provide evidence that rhipidistian crossopterygian lived in river channels and lakes on large deltas.
B. To identify an effect of the oxidation of iron minerals on the evolution of rhipidistian crossopterygian.
C. To help explain why scientists have concluded that rhipidistian crossopterygian probably had lungs.
D. To explain why scientists decided to cut cross sections through some fossils of rhipidistian crossopterygian.

5. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
A. Because the lungs of these fish were able to provide only a small amount of oxygen, these fish obtained most of their oxygen through their gills during periods of drought.
B. During periods of extended drought, these fish used their lungs to increase their intake of oxygen beyond the levels absorbed by the gills in normal times.
C. Although these fish primarily used their gills to obtain oxygen, they used their lungs to obtain oxygen from the air when there was not enough in the water.
D. During periods of extended drought, the gills became an auxiliary breathing device and the lungs became the main source of oxygen for these fish.

6. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
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B. During periods of extended drought, these fish used their lungs to increase their intake of oxygen beyond the levels absorbed by the gills in normal times.
C. Although these fish primarily used their gills to obtain oxygen, they used their lungs to obtain oxygen from the air when there was not enough in the water.
D. During periods of extended drought, the gills became an auxiliary breathing device and the lungs became the main source of oxygen for these fish.

7. The word "adjacent" in the passage is closest in meaning to:
A. nearby
B. available
C. temporary
D. fresh

8. The word "progressively" in the passage is closest in meaning to:
A. increasingly
B. noticeably
C. occasionally
D. rapidly

9. In paragraph 4, why does the author point out that crabs and other arthropods were already living on land when the ancestors of the first tetrapods began living there?
A. To account for the presence of dead fish along the edges of ponds and streams during the Devonian.
B. To support the claim that climate change caused freshwater habitats to become more restricted during the Devonian.
C. To identify a consequence of the emergence of plants into terrestrial habitats near ponds and streams.
D. To identify a possible reason for why certain fish gradually became terrestrial organisms.

10. According to paragraph 4， teeth of the earliest tetrapods suggest that these tetrapods
A. competed with other animals for protein
B. were probably carnivores
C. could easily climb out of water
D. were able to eat plants

11. According to paragraph 5, which of the following was true of the first tetrapods?
A. They became dependent for food on organisms already living on land.
B. They needed to develop new mechanisms for obtaining nutrients.
C. They continued to live in close association with aquatic environments.
D. They were evolutionarily far removed from their rhipidistian ancestors.

12. According to paragraph 5, what was the main way that the earliest tetrapods differed from their immediate fish ancestors?
A. The tetrapods had a different skeletal structure.
B. The tetrapods had more sources of food available
C. The tetrapods had a circulation system.
D. The tetrapods could move to new pools of water.

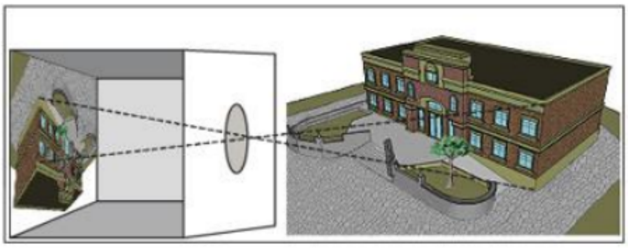
13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.
These would have been deposited by the receding waters of droughts, during which many aquatic animals must have died.
Where would the sentence best fit? Click on a square [■] to add the sentence to the passage

14. **Directions:** An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. **This question is worth 2 points.** Drag your answer choices to the spaces where they belong. To remove an answer choice, click on it. To review the passage, click VIEW TEXT

Freshwater lobe-finned fish may be the direct ancestors of terrestrial tetrapods.

A. Rhipidistian crossopterygian had features such as primitive lungs and thick fins that could have helped it survive dry periods.
B. During the Devonian, the number of bones increased in the fins of rtiipidistians, improving such animals' ability to swim and move over land
C. Shortly after the earliest tetrapods developed lungs, plants and other animals began to flourish on land.
D. By the Devonian period, lobe-finned fish preferred freshwater habitats to life in the ocean.
E. A drier climate and new sources of food on land may have encouraged the lobe-finned fish's move to a terrestrial existence.
F. Early tetrapods remained closely connected to water, but several of their body structures were adapted for life on land.

2. The Use of the Camera Obscura



The precursor of the modern camera, the camera obscura is a darkened enclosure into which light is admitted through a lens in a small hole. The image of the illuminated area outside the enclosure is thrown upside down as if by magic onto a surface in the darkened enclosure. This technique was known as long ago as the fifth century B.C. in China. Aristotle also experimented with it in the fourth century B.C., and Leonardo da Vinci described it in his notebooks in 1490. In 1558 Giovanni Battista Della Porta wrote in his twenty-volume work ***Magia naturalis*** (meaning "natural magic") instructions for adding a convex lens to improve the quality of the image thrown against a canvas or panel in the darkened area where its outlines could be traced. Later, portable camera obscuras were developed, with interior mirrors and drawing tables on which the artist could trace the image. For the artist, this technique allows forms and linear perspective to be drawn precisely as they would be seen from a single viewpoint. Mirrors were also used to reverse the projected images to their original positions.

Did some of the great masters of painting, then, trace their images using a camera obscura? Some ait historians are now looking for clues of artists' use of such devices. One of the artists whose paintings are being analyzed from this point of view is the great Dutch master, Jan Vermeer, who lived from 1632 to 1675 during the flowering of art and science in the Netherlands, including the science of optics. Vermeer produced only about 30 known paintings, including his famous ***The Art of Painting*.** The room shown in it closely resembles the room in other Vermeer paintings, with lighting coming from a window on the left, the same roof beams, and similar floor tiles, suggesting that the room was fitted with a camera obscura on the side in the foreground. The map hung on the opposite wall was a real map in Vermeers possession, reproduced in such faithful detail that some kind of tracery is suspected. When one of Vermeer's paintings was X-rayed, it did not have any preliminary sketches on the canvas beneath the paint, but rather the complete image drawn in black and white without any trial sketches. Vermeer did not have any students, did not keep any records, and did not encourage anyone to visit his studio, facts that can be interpreted as protecting his secret use of a camera obscura.

In recent times the British artist David Hockney has published his investigations into the secret use of the camera obscura, claiming that for up to 400 years, many of Western art's great masters probably used the device to produce almost photographically realistic details in their paintings. He includes in this group Caravaggio, Hans Holbein, Leonardo da Vinci, Diego Velazquez, Jean-Auguste-Dominique Ingres, Agnolo Bronzino, and Jan van Eyck. From an artist's point of view, Hockney observed that a camera obscura compresses the complicated forms of a three-dimensional scene into two-dimensional shapes that can easily be traced and also increases the contrast between light and dark, leading to the chiaroscuro effect seen in many of these paintings. In Jan van Eyck's *The Marriage of Giovanni Arnolfini and Giovanna Cenami,* the complicated foreshortening in the chandelier and the intricate detail in the bride's garments are among the clues that Hockney thinks point to the use of the camera obscura.

So what are we to conclude? If these artists did use a camera obscura, does that diminish their stature? Hockney argues that the camera obscura does not replace artistic skill in drawing and painting. In experimenting with it, he found that it is actually quite difficult to use for drawing, and he speculates that the artists probably combined their observations from life with tracing of shapes.

1. What can be inferred from paragraph 1 about Giovanni Battista Della Porta's contribution to the camera obscura?
A. He translated a Chinese description of the use of the camera obscura and made the technique available to artists.
B. His convex lens made the projected image easier to trace.
C. His version of the camera obscura allowed for the later addition of a mirror.
D. His improvements relied heavily on design changes proposed earlier by Leonardo da Vinci.

2. The word "portable" in the passage is closest in meaning to
A. valuable
B. practical
C. moveable
D. popular

3. The word "projected" in the passage is closest in meaning to
A. whole
B. corrected
C. enlarged
D. shown

4. Paragraph 2 answers which of the following questions about paintings by Vermeer?
A. What characteristics of Vermeer's paintings suggest that he may have used a camera obscura?
B. Why did Vermeer produce only about 30 paintings?
C. Do Vermeer's paintings in general suggest that he was unable to paint accurately without using a camera obscura?
D. Why did Vermeer need to draw an image on the canvas of the painting that was X-rayed if he was using a camera obscura?

5. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
A. One artist with a particularly interesting point of view about the use of the camera obscura in painting was Jan Vermeer, who lived in the Netherlands from 1632 to 1675.
B. Historical analyses suggest that Dutch masters were interested in the science of optics, so they may have used the camera obscura to trace images.
C. The use of the camera obscura is being analyzed in the paintings of Jan Vermeer, who lived in the Netherlands when art and science were flourishing there.
D. One view held by historians is that most Dutch masters were as interested in art as they were in

science, and that provides clues about the techniques used in their paintings.

6. What does paragraph 2 indicate about Vermeer's *The Art of Painting* ?
A. It is the first in a series of about 30 paintings that he created.
B. It may have been painted by one of his students.
C. it was in his possession until his death in 1675.
D. It has the same setting as several other works of his.

7. The word "faithful" in the passage is closest in meaning to
A. unusual
B. extensive
C. exact
D. historical

8. Why does the author provide the information that 'When one of Vermeer's paintings was X-rayed, it did not have any preliminary sketches on the canvas beneath the paint, but rather the complete image drawn in black and white without any trial sketches"?
A. To provide an example of a way to learn about the practices of artists who did not keep good records
B. To emphasize Vermeer's confidence and skill as an artist
C. To provide evidence that Vermeer may have traced the image using a camera obscura
D. To argue that Vermeer did his preliminary sketching on paper, rather than on canvas

9. According to paragraph 3, Hockney believes that all of the following indicate use of a camera obscura EXCEPT
A. very detailed, realistic work
B. increased contrast between light and dark
C. oversimplification of forms when the image is traced
D. complicated foreshortening of objects

10. The word "ntricate" in the passage is closest in meaning to
A. surprising
B. complex
C. beautiful
D. clear

11. The word "diminish" in the passage is closest in meaning to
A. reduce
B. affect
C. reflect
D. determine

12. According to paragraph 4， what does Hockney argue about the use of the camera obscura in producing art?
A. Works produced using a camera obscura do not deserve as much respect as those produced without it.
B. The camera obscura was probably used primarily as a training device, rather than used in producing finished works.
C. Use of the camera obscura by Western art's great masters was probably relatively rare.
D. While the use of the camera obscura may have helped artists, they still needed to have significant artistic ability.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.
All these developments helped artists to create accurate images of objects, people, and scenes.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage

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The camera obscura is a darkened enclosure into which light is admitted through a lens in a small hole

A. Evidence that the use of the camera obscura has long been known is provided by its description in many sources, including works dating back to Chinese writers from the fifth century B.C.
B. Some historians who have studied paintings by Western masters have found clues indicating that the masters may have secretly used the camera obscura in their works.
C. It is now widely believed that the use of the camera obscura led to the development of a style of photographic realism in Western art.
D. The camera obscura was most widely used by artists in Seventeenth-century Netherlands, a period when art and science thrived
E. The unique features of Vermeer's *The Art of Painting* make it unlikely that it was made with a camera obscura, as opposed to his other works.
F. The artist David Hockney has speculated that artists probably combined the use of the camera A with their own original observations from life.

3. Seagrasses

Many areas of the shallow sea bottom are covered with a lush growth of aquatic flowering plants adapted to live submerged in seawater. These plants are collectively called seagrasses. Seagrass beds are strongly influenced by several physical factors. The most significant is water motion: currents and waves. Since seagrass systems exist in both sheltered and relatively open areas, they are subject to differing amounts of water motion. For any given seagrass system, however, the water motion is relatively constant. Seagrass meadows in relatively turbulent waters tend to form a mosaic of individual mounds, whereas meadows in relatively calm waters tend to form flat, extensive carpets. The seagrass beds, in turn, dampen wave action, particularly if the blades reach the water surface. This damping effect can be significant to the point where just one meter into a seagrass bed the wave motion can be reduced to zero. Currents are also slowed as they move into the bed.

The slowing of wave action and currents means that seagrass beds tend to accumulate sediment. However, this is not universal and depends on the currents under which the bed exists. Seagrass beds under the influence of strong currents tend to have many of the lighter particles, including seagrass debris, moved out, whereas beds in weak current areas accumulate lighter detrital material. It is interesting that temperate seagrass beds accumulate sediments from sources outside the beds, whereas tropical seagrass beds derive most of their sediments from within.

Since most seagrass systems are depositional environments, they eventually accumulate organic material that leads to the creation of fine-grained sediments with a much higher organic content than that of the surrounding unvegetated areas. This accumulation, in turn, reduces the water movement and the oxygen supply. The high rate of metabolism (the processing of energy for survival) of the microorganisms in the sediments causes sediments to be anaerobic (without oxygen) below the first few millimeters. According to ecologist J. W. Kenworthy, anaerobic processes of the microorganisms in the sediment are an important mechanism for regenerating and recycling nutrients and carbon, ensuring the high rates of productivity—that is, the amount of organic material produced-that are measured in those beds. In contrast to other productivity in the ocean, which is confined to various species of algae and bacteria dependent on nutrient concentrations in the water column, seagrasses are rooted plants that absorb nutrients from the sediment or substrate. They are, therefore, capable of recycling nutrients into the ecosystem that would otherwise be trapped in the bottom and rendered unavailable.

Other physical factors that have an effect on seagrass beds include light, temperature, and desiccation (drying out). For example, water depth and turbidity (density of particles in the water) together or separately control the amount of light available to the plants and the depth to which the seagrasses may extend. Although marine botanist W. A. Setchell suggested early on that temperature was critical to the growth and reproduction of eelgrass, it has since been shown that this particularly widespread seagrass grows and reproduces at temperatures between 2 and 4 degrees Celsius in the Arctic and at temperatures up to 28 degrees Celsius on the northeastern coast of the United States. Still, extreme temperatures, in combination with other factors, may have dramatic detrimental effects. For example, in areas of the cold North Atlantic, ice may form in winter. Researchers Robertson and Mann note that when the ice begins to break up, the wind and tides may move the ice around, scouring the bottom and uprooting the eelgrass. In contrast, at the southern end of the eelgrass range, on the southeastern coast of the United States, temperatures over 30 degrees Celsius in summer cause excessive mortality. Seagrass beds also decline if they are subjected to too much exposure to the air. The effect of desiccation is often difficult to separate from the effect of temperature. Most seagrass beds seem tolerant of considerable changes in salinity (salt levels) and can be found in brackish (somewhat salty) waters as well as in full- strength seawater.

1. According to paragraph 1, which of the following is true about seagrasses in calm ocean waters?
A. They will not survive for very long without the nutrients brought In by fast-moving waters.
B. They tend to form beds covering large areas along the ocean floor.
C. They usually are arranged in separate mounds.
D. They grow more slowly than do seagrasses in fast-moving waters.

2. According to paragraph 1, which of the following is MOST likely to describe a bed in which seagrasses reach the surface of the water?
A. The water is almost completely still.
B. The bed often has major damage from strong waves or currents.
C. The bed is generally no more than one square meter in size.
D. Grasses form a mosaic of individual mounds.

3. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.
A. Light particles and debris collect in some seagrass beds, but are washed out of those affected by strong currents.
B. Seagrass beds under the influence of strong currents tend to accumulate many of the lighter particles from other beds
C. The strength of the currents determines how quickly accumulated seagrass debris is moved out of the beds.

D. Seagrass debris and other light particles are often moved from areas of strong currents to areas of weak currents.

4. The word "derive" in the passage is closest in meaning to
A. maintain
B. expel
C. obtain
D. enrich

5. According to paragraph 3, which of the following does NOT accurately describe the sediments that collect in seagrass beds?
A. Fine-grained
B. Only a few millimeters deep
C. Low in oxygen
D. Rich in organic matter

6. The word "confined" in the passage is closest in meaning to
A. related
B. limited
C. relevant
D. helpful

7. According to paragraph 3，how do seagrasses affect the nutrient supply in the ecosystem?
A. Because of their high rate of metabolism, they consume a large percentage of the available nutrients.
B. They attract various species of algae and bacteria that produce high nutrient concentrations in the water column.
C. They take up carbon and other nutrients trapped on the sea bottom and bring them back into use.
D. Through anaerobic processes at their roots, they produce a very nutrient-rich sediment.

8. It can be inferred from paragraph 4 that the reason seagrasses do not grow in very deep water is that
A. they cannot handle intense water pressure
B. deep water is too cold
C. they would not get enough light
D. deep water is too salty

9. The word “detrimental” in the passage is closest in meaning to
A. harmful
B. significant
C. unexpected
D. distinct

10. The word “detrimental” in the passage is closest in meaning to
A. harmful
B. significant
C. unexpected
D. distinct

11. Paragraph 4 suggests that which of the following would be the LEAST likely to cause major damage to eelgrass and other common seagrasses?
A. Factors related to extreme temperatures
B. Exposure to air
C. Major changes in salinity
D. The movement of ice on the seafloor

12. The phrase “tolerant of” in the passage is closest in meaning to
A. unused to
B. strongly affected by
C. protected from
D. able to withstand

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.
Seagrasses grow together in dense patches, or beds, with as many as 4,000 blades per square meter.
Where would the sentence best fit? Click on a square [■] to add the sentence to the passage

14. Look at the four squares [■] that indicate where the following sentence could be added to the passage.
Seagrasses grow together in dense patches, or beds, with as many as 4,000 blades per square meter.
Where would the sentence best fit? Click on a square [■] to add the sentence to the passage
Drag your answer choices to the spaces where they belong. To remove an answer choice, click on it.
To review the passage, click **VIEW TEXT**

Seagrasses are aquatic flowering plants that grow in either sheltered or open areas of the sea

A. Seagrass beds are influenced by several physical factors, the most significant being the stability of the sea bottom, which must anchor them against the currents.
B. Because they slow currents and waves, seagrass beds collect deposits of rich organic sediments, which are home to many anaerobic microorganisms.
C. Unlike sea organisms that depend on the water column for their productivity, seagrasses ensure high rates of productivity by taking nutrients from ocean floor sediment.
D. Sediments in seagrass beds vary by region, with temperate beds accumulating sediments from within, and tropical beds collecting sediments from without.
E. Seagrasses under weak currents tend to have higher rates of metabolism than those under strong currents, perhaps because of differences in oxygen levels.
F. Although seagrasses survive in temperatures ranging from 2 to 28 degrees Celsius, more extreme temperatures can damage them, as can desiccation and lack of light.

[Listening]

Conversation 1
1. Why does the student go to see the professor?
A. To get his opinion about why a project she recently completed had unexpected results.
B. To discuss how a topic covered in class is similar to her group’s research topic.
C. To ask him for suggestions to address a problem in her research.
D. To discuss the professor's concern about her group's research project.

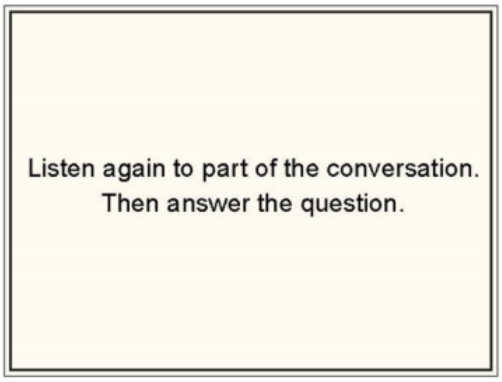
2. In response to the professor’s question, what does the woman say about Tom and Jane?
A. They are working on an assignment for another class.
B. They are already observing students for the research project.
C. They are dealing with a technical issue at the library.
D. They are making arrangements at the library for their research project.

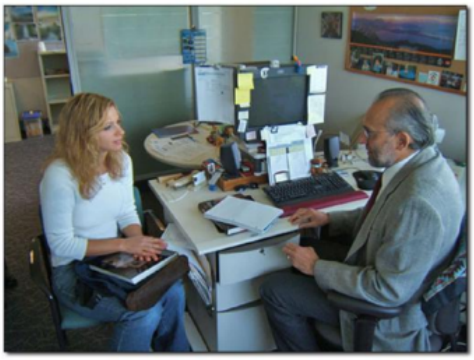
3. What had the group of students planned to research?
A. The effect of noise on the productivity of library employees.

B. The effect of changing the amount of light in the library.
C. The study habits of students in the library.
D. The effect that being observed while studying has on students.

4. Why does the professor mention lighting?
A. To explain why production costs gradually increased over the years at a manufacturing plant
B. To give a reason for a decline in productivity at two manufacturing plants
C. To compare the working conditions at two manufacturing plants
D. To give an example of a working condition that was adjusted at a manufacturing plant

5. Why does the student say this?





A. She is disappointed with the observations that the members of her group have made so far.
B. She does not understand the point that the professor is making.
C. She wants to determine a way for her group to make observations in secret.
D. She is aware that her group's presence might affect student behavior.

Lecture 1
1. What is the main purpose of the lecture?
A. To provide an example of a practical use of nanotechnology.
B. To show the origins of the field of nanotechnology.
C. To give a brief outline of the main concepts of nanotechnology.
D. To explain the growing interest in nanotechnology research.

2. How does the professor organize the information he presents to the class?
A. He describes the inspiration behind the nanocoating, then how the coating works.
B. He describes how the nanocoating is currently marketed, then the inspiration behind it.
C. He explains how fogging occurs, then the basic concepts of nanotechnology.
D. He explains how fogging occurs, then how the nanocoating prevents it.

3. According to the professor, how does the new nanocoating work?
A. By forcing light to bounce off a glass-coated polymer.
B. By forcing water droplets to roll off an ultrathin surface.
C. By causing water droplets to merge into a single sheet of water.
D. By causing light to scatter randomly in many directions.

4. According to the information in the lecture, why does the new nanocoating not last as long on plastic as it does on glass?
A. Plastic cannot withstand extremely high temperatures.

B. The internal structure of plastic repels a positively charged polymer.
C. The coating solution scatters when it comes into contact with plastic.
D. Plastic surfaces scratch more easily than glass surfaces do.

5. What inspired the team of scientists in developing the new coating?
A. A problem the team frequently encountered in everyday life.
B. The ineffectiveness of spray solutions in flattening water droplets.
C. The leaves of a plant that the team had been investigating.
D. Interactions observed between silica nanoparticles and polymers.

6. What is the professor's opinion about the approach inventors took to the development of the new nanocoating?
A. He thinks other inventors should use a similar approach.
B. He is impressed by the flexibility of their approach.
C. He is surprised the research process took so long.
D. He thinks they should have spent more time testing a superhydrophobic coating.

Lecture 2

1. What is the lecture mainly about?
A. The dramatic structure of an ancient Greek play.
B. The influence of ancient Greek theater design on modern theaters.
C. The design of ancient Greek theaters.
D. The role of plays in ancient Greek society.

2. What were two purposes served by the skene? Click on 2 answers.
☐ It provided extra seating for the audience.
☐ It served as the location for the play’s action.
☐ It was used to store items needed for the play.
☐ It was where the chorus performed.

3. Why is the play Hippolytus discussed?
A. To give an example of a strategy used by ancient Greek playwrights.
B. To give an example of how animals were used in ancient Greek plays.
C. To identify the first use of a messenger in an ancient Greek play.
D. To point out that ancient Greek plays typically involved tragic events.

4. What point does the professor make about the chorus in ancient Greek plays?
A. It performed only at the beginning of the play.
B. It interpreted what was happening on the stage.
C. It did relatively little singing and dancing.
D. it was less important than it is in modern plays.

5. How did Aristotle view the chorus?
A. As the author of the play.
B. As a distraction from the story.
C. As a messenger reporting news.
D. As a character in the play.

6. Why does the professor say this:
A. To emphasize the popularity of plays in ancient Greece.
B. To praise the creativity of the ancient Greeks.
C. To point out that every seat in an ancient Greek theater had a clear view of the stage.

D. To suggest that audiences in ancient Greece could hear plays better than they could see them.

Conversation 2

1. Why does the student go to see the professor?
A. To find out whether the university gallery accepts student artwork.
B. To find out how artwork is selected for university gallery exhibitions.
C. To learn the focus of an upcoming exhibition of student artwork.
D. To learn which techniques the professor will cover in his class on abstract art.

2. Why does the student mention Jackson Pollock?
A. To indicate to the professor that she is familiar with the drip technique.
B. To find out if she can see an original painting by Jackson Pollock in the university gallery.
C. To make a comparison between her paintings and those of Jackson Pollock.
D. To express interest in taking the professor's class.

3. According to the professor, what distinguishes Jackson Pollock's work?
A. The method he used for applying paint to a canvas.
B. The location where he did his work.
C. The widespread popularity of his style of art.
D. The size of the paintings he produced.

4. What does the professor imply about his class on abstract art?
A. It focuses primarily on Jackson Pollock.
B. It is one of the most popular classes in the department.
C. It tends to attract the department's best students.
D. It encourages students to explore different painting techniques.

5. What does the professor imply when he says this:
A. Students must have taken a course in the technique featured in the exhibition.
B. Exhibiting in the university gallery is a degree requirement for art majors.
C. Artwork submitted by a first-year student will probably not be accepted.
D. The woman should submit her painting soon because the deadline is approaching.

Lecture 3

1. What is the lecture mainly about?
A. The spread of early agricultural methods from New Guinea to other cultures.
B. Differences in the types of crops grown in early centers of agriculture.
C. Evidence supporting the theory that agriculture developed independently in New Guinea.
D. Techniques used by researchers to identify farming methods in the earliest centers of agriculture.

2. According to the professor, why was the archaeological evidence found in New Guinea during the 1960s and 1970s inconclusive? Click on 2 answers.
☐ Construction of agricultural drainage ditches had damaged much of the archaeological evidence.
☐ Plant remains were not well preserved in the climate of New Guinea.
☐ Ancient types of domestic plants were no longer grown by modern farmers.
☐ It was unclear whether evidence of early deforestation suggested planting or hunting.

3. Why does the professor talk about layers of soil?
A. To show how phases of agricultural development were linked to evidence of population growth.
B. To describe how researchers identified several phases of agricultural development in New

Guinea.
C. To illustrate how swampy conditions cause archaeological remains to deteriorate.
D. To describe one of the methods of farming used at Kuk.

4. Why did researchers conclude that the taro remains found near Kuk were a result of farming?
A. Taro does not grow wild in highland areas.
B. Taro is a native plant of China.
C. Taro was only found in a very small area near Kuk.
D. Taro normally does not grow in wet climates.

5. What evidence indicated that bananas were being cultivated in New Guinea during an earlier period than was previously thought? Click on 3 answers.
☐ High concentrations of fossil remains of bananas.
☐ The discovery of stone tools designed to harvest bananas.
☐ The presence of regularly distributed mounds.
☐ Indications that Kuk did not become a swampy wetland until approximately 7,000 years ago.
☐ Genetic analyses of banana remains in New Guinea and Southeast Asia.

6. What point does the professor make about the theory that agriculture brings about social change?
A. Recent research has yielded unexpected evidence supporting the theory.
B. The theory seems to be contradicted by the development of society in New Guinea.
C. Future discoveries in Kuk are likely to provide definitive proof for the theory.
D. The theory explains why New Guinea has become an egalitarian society.

Lecture 4

1. What is the main purpose of the talk?
A. To explain the mechanical functioning of barrages.
B. To discuss some possible ecological effects of building barrages.
C. To discuss the effects of ocean tides on coastal ecosystems.
D. To describe ways to increase biological productivity of estuaries.

2. What is the professor's opinion when the man expresses concern about marine life on the mud flats?
A. She thinks the environmental impact of barrages would be worse for birds than for fish.
B. She agrees that any damage to the mud flats would probably be irreversible.
C. She feels that the situation is more complex than the man realizes.
D. She does not believe that the mud flats support a wide variety of animal species.

3. The professor mentions a change in the quantity of fish caught near a barrage in France. What most likely happened to the water in the estuary as a result of the construction of the barrage?
A. The water has become clearer.
B. The water has become saltier.
C. The water has become polluted
D. The water level has risen.

4. What does the professor say is a criticism of the proposed barrage at the Severn River in Great Britain?
A. It would damage nearby buildings.
B. It would attract harmful species to the region.
C. It would not be as large as the barrage in France.
D. it would be too expensive to build.

5. Why does the professor say this:
A. To request that the student answer in greater detail.
B. To introduce a new topic for discussion.
C. To make the student's statement more accurate.
D. To repeat a point that she had not stated clearly.

[Speaking]

Task 1.

Talk about a special job you have had in the past or would like to have in the future.

Task 2.

Some people believe that primary schools should no longer teach children how to write by hand, and instead should spend time teaching them how to type on a computer. Other people believe that it is still important for schools to teach children to have good handwriting. Which point of view do you agree with? Explain why.

Task 3.

Reading Time: 50 seconds

Create Student Committee to Decide Funding for Student Organizations

I think students should be in charge of deciding which student organizations (for example, the jazz band or the hiking club) receive money from the university. Students should also be in charge of how much money each organization receives. A special committee made up of students could be created to make these decisions. Currently, these funding decisions are made by university administrators, but a student committee would know better than the administrators which organizations are most important to students and most deserving of the university's financial support. I'm sure a lot of students would be interested in serving on the committee, and those who do serve will gain valuable leadership experience.

Q: The man expresses his opinion of the letter writer's proposal. State his opinion and explain the reasons he gives for holding that opinion.

Task 4.

Reading Time: 50 seconds

Scope Creep

Businesses that perform services or carry out projects for clients generally come to an agreement with their clients about the extent or scope of a project before beginning the project. However, as a project progresses, clients may ask for more than the business originally expected to provide, and the scope of the project may grow larger than intended This phenomenon is known as scope creep, and it can cause conflict between businesses and their clients. Scope creep is especially common when the terms or conditions of the initial agreement are not clearly defined, and a client may expect more than the business had planned to provide.

Q: Explain how the example in the lecture illustrates the concept of scope creep.

Task 5

Briefly summarize the problem the speaker are discussing. Then state which solution you would recommend. Explain the reasons for your recommendation.

Task 6.

Using the examples of wild turkeys and beetles, explain two benefits of forest fires for animals.

[Writing]

1.

In 1957 a European silver coin dating to the eleventh century was discovered at a Native American archaeological site in the state of Maine in the United States. Many people believed the coin had been originally brought to North America by European explorers known as the Norse, who traveled across the Atlantic Ocean and came into contact with Native Americans almost a thousand years ago.

However, some archaeologists believe that the coin is not a genuine piece of historical evidence but a historical fake; they think that the coin was placed at the site recently by someone who wanted to mislead the public. There are three main reasons why some archaeologists believe that the coin is not genuine historical evidence.

Great Distance from Norse Settlements
First, the Native American site in Maine where the coin was discovered is located very far from other sites documenting a Norse presence in North America. Remains of Norse settlements have been discovered in far eastern Canada. The distance between the Maine site and the Norse settlements in Canada is more than a thousand kilometers, suggesting the coin has no real connection with the settlements.

No Other Coins Found
A second problem is that no other coins have been found at the Canadian sites that were inhabited by the Norse. This suggests that the Norse did not bring any silver coins with them to their North American settlements.

Third, the Norse who traveled to North America would have understood that silver coins would most likely be useless to them. Sitver coins may have been in wide use in Europe at the time, but the Norse, as experienced explorers, would have known that native North Americans did not recognize silver coins as money.

1. Directions: You have 20 minutes to plan and write your response. Your response will be judged on the basis of the quality of your writing and on how well your response presents the points in the lecture and their relations to the reading passage.

Questions: Summarize the points made in the lecture, being sure to explain how they challenge the specific theories presented in the reading passage.

2. Directions: Read the question below. You have 30 minutes to plan, write, and revise your essay. Typically, an effective response will contain a minimum of 300 words.

Question:
Some people believe that when busy parents do not have a lot of time to spend with their children, the best use of that time is to have fun playing games or sports. Others believe that it is best to use that time doing things together that are related to schoolwork. Which of the two approaches do you prefer?

Use specific reasons and examples to support your answer.